

# Syarfandi "Andi" Achmad

Data Scientist | ML Engineering & Predictive Analytics Specialist Makassar, Indonesia | [+62 852-5652-2335](tel:+6285256522335) | [syarfandi.achmad@gmail.com](mailto:syarfandi.achmad@gmail.com) | Portfolio: [syarfandi.github.io](https://syarfandi.github.io) | [LinkedIn](#)

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## Professional Summary

Insightful **Data Scientist** with 4+ years extracting actionable intelligence from complex datasets. Expert in **Python, Machine Learning, Statistical Modeling**, and **Predictive Analytics** using real-world production data. Proven track record delivering data-driven solutions for government (9M+ citizen records) and enterprise retail (100+ module POS). Skilled at building ML pipelines, interactive dashboards, and AI-powered applications.

## Core Competencies

- **ML/AI:** Scikit-Learn, TensorFlow, ARIMA/Prophet, K-Means, DBSCAN, NLP, LLM Integration (Gemini AI), Anomaly Detection.
- **Python:** Pandas, NumPy, SciPy, Matplotlib, Seaborn, Jupyter Notebooks, Feature Engineering.
- **SQL & Data:** PostgreSQL (Stored Procedures, Aggregation), Supabase, ETL Pipelines, Data Warehousing, Data Cleaning.
- **Visualization:** Matplotlib, Seaborn, D3.js, Interactive Dashboards, Geospatial Visualization.
- **Statistics:** Hypothesis Testing, Regression, Classification, A/B Testing, Time-Series Analysis, Bayesian Methods.
- **Tools:** Git, Docker, GitHub Actions, Jupyter, Google Colab, Linux.

## Projects

- [Kawan Sedarah \(Blood Stock\)](#): A **command center** dashboard that tracks **blood stock** availability across donor units in **real-time**, powered by **Supabase** as its core backend.
- [Portal Data Sulsel](#): A unified information system delivering **data visualization, monitoring**, and religious **statistical analysis** across all districts in South Sulawesi in **real-time**.
- [AI Virtual Assistant](#): A **Conversational AI**-powered virtual assistant featuring **Realtime Lipsync** on 3D avatars, backed by **Voice AI** and **LLM** for immersive conversational interactions.
- [GetKasir Enterprise POS](#): A large-scale **Enterprise POS** ecosystem with **100+ modules**, integrating **AI**-driven sales prediction, **biometric** authentication, and a scalable **cloud-native** architecture.
- [PPID Kemenag Kota Parepare](#): A public information disclosure portal for **PPID Kemenag** providing access to **official documents, performance reports**, and an intelligent chatbot powered by **Google Gemini AI**.
- [OP Tracker](#): A premium logbook for tracking **One Piece** watch progress without login — equipped with automatic **filler filter**, **saga & arc** navigation map, and a gamified **pirate rank** system.
- [QR Scanner Presence App](#): A modern **attendance solution** leveraging **QR Code** scanning for instant participant check-in, built on a **Next.js + Supabase** stack with **real-time** sync.
- [Malmora \(Reseller & Dropship\)](#): A **reseller & dropship** marketplace tailored for Muslim lifestyle products, enabling anyone to sell without physical stock across **Sulawesi and Eastern Indonesia**.
- [Email Server Makassar Gov](#): A city-scale **Mailcow**-based email cluster serving **200+ official domains** for Makassar City Government, hardened with **SPF/DKIM/DMARC** security protocols and **high-availability** setup.
- [Helpdesk Kota Makassar](#): An **e-government** system for handling **citizen complaints** digitally — featuring **real-time ticket tracking**, automated escalation, and cross-department integration

across **Makassar City agencies**.

- **Virtual Museum Makassar:** An immersive digital museum experience with **360° panoramic** tours, interactive **3D artifact** collections, and **audio-visual** narratives preserving South Sulawesi cultural heritage.
  - **AI-Haram VR:** A high-precision **virtual reality** simulation of **Al-Masjid Al-Haram** and **Al-Masjid An-Nabawi**, developed in **Unity** and optimized for both **VR headsets** and mobile devices as a Hajj & Umrah educational tool.
  - **Paralluta (Lalu Lintas):** A smart platform for **traffic management** in Makassar City — covering **road condition monitoring**, violation logging, and centralized **field officer coordination**.
  - **SOP Pelaporan Insiden Siber:** A **SOP** framework and **cyber incident response protocols** designed for **Makassar City Government** IT infrastructure, aligned with **BSSN** and **NIST Cybersecurity Framework** standards.
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## Experience

### Computer Specialist (Civil Servant) | Ministry of Religious Affairs

2025 – Present | Makassar

- **Built** the **Data Sulsel** analytics engine — PostgreSQL aggregation pipelines, statistical dashboards, and trend analysis for 9M+ citizen service records.
- **Developed** predictive models for resource allocation, using historical data to forecast demand across 24 regencies.
- **Integrated Google Gemini AI** into public portals (PPID), implementing NLP-powered document search and automated response generation.

### DevOps & Infrastructure Expert | Diskominfo Makassar

2023 – 2025 | Makassar

- **Built** sentiment analysis models for **Helpdesk Makassar** complaint data — NLP classification to auto-prioritize critical urban issues.
- **Developed** anomaly detection algorithms for government email clusters, identifying patterns across 200+ domains.

### Co-Founder | GetKasir (Enterprise POS)

2019 – Present | Remote

- **Built** sales forecasting models using time-series (ARIMA/Prophet) for **100+ module POS**, improving inventory accuracy by 40%.
- **Developed** merchant clustering (K-Means/DBSCAN) for churn risk identification and targeted marketing segments.
- **Designed** automated ETL pipelines for high-volume transaction data — cleaning, feature engineering, aggregation.

### SQA & DevOps Engineer | Malmora (Startup)

2020 – 2023 | Remote

- **Analyzed** sales data and purchase patterns to optimize product ranking and recommendation algorithms.
- **Built** A/B testing frameworks for conversion optimization, analyzing statistical significance on order completion.

- **Developed** regional market dashboards tracking seller performance, demand trends, and logistics metrics.
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## Awards

- **National Cyber Exercise #10** – BSSN (2024)
- **Top 10** – Samsung Next Apps 4.0 (2017)
- **Microsoft CityApp** – Participant

## Education

- **Bachelor of Computer Science (S.Kom)** | UIN Alauddin Makassar *GPA: 3.22 / 4.00*
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## Data Scientist Expertise Keywords

Python, SQL, Machine Learning, Deep Learning, Scikit-Learn, TensorFlow, PyTorch, Pandas, NumPy, NLP, LLMs, Computer Vision, Predictive Modeling, Statistics, Time Series Forecasting, A/B Testing, Feature Engineering, Model Deployment, Jupyter, MLOps.

## Future-Ready Technical Roadmap

To stay competitive in the global Data & Analytics market, I am actively implementing:

- **Predictive Analytics:** Implementing advanced ML models for forecasting and trend analysis.
- **Real-time Streaming:** Building low-latency data pipelines using Apache Flink and Spark Streaming.
- **Vector Databases for LLMs:** Architecting data storage for large language model retrieval (RAG).
- **Automated Data Governance:** Ensuring data compliance and quality through automated AI-driven auditing.